

ESMCA: Explanation-Steered Modular Continual Adapters

Research Proposal

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1 Problem Statement

Continual learning (CL) systems must sequentially acquire new tasks without erasing previously learned knowledge — the well-known *catastrophic forgetting* problem [1]. Low-Rank Adaptation (LoRA) [2] has emerged as a parameter-efficient fine-tuning paradigm that partially mitigates forgetting by training task-specific low-rank adapters on top of a frozen pretrained backbone. However, existing LoRA-based CL methods require either:

- (a) **Task labels at inference** — impractical in real deployments where the task identity is unknown, or
- (b) **A learned black-box router** — which itself can be a forgetting bottleneck and offers no interpretability.

Central Research Question: Can *attribution maps* — signals from explainability methods — replace learned task routers in a LoRA continual learning system, eliminating the need for task labels while simultaneously providing an interpretable, proactive forgetting detector?

2 Novelty Analysis and Literature Gap

2.1 What Already Exists

Table 1: Prior work and their limitations relevant to ESMCA.

Method	Key Idea	Limitation
C-LoRA	Orthogonal routing matrix per task	Router is a black-box weight
AM-LoRA	Attention-weighted adapter routing	Routing learned from task loss only
L2R	Output-space adapter composition	Requires domain/task identifiers
CABLE	Gradient-similarity routing	Task labels required
XAI + CL	Explanation drift for regularization	Not used as primary routing signal
FADMON	Attribution drift for RL monitoring	Monitoring only; not applied to CL adapters

2.2 The Untouched Gap

No existing work uses attribution maps as a **dual-purpose signal** — routing adapters at inference *and* detecting forgetting at training time — within a LoRA-based continual learning framework. This is the novel core of ESMCA.

3 Architecture: ESMCA

3.1 System Overview

Figure 1 illustrates the full ESMCA pipeline. At its core, ESMCA separates the standard CL pipeline into five interacting modules: a frozen backbone, an attribution map extractor, an attribution-based router, a task adapter bank, and an explanation drift monitor.

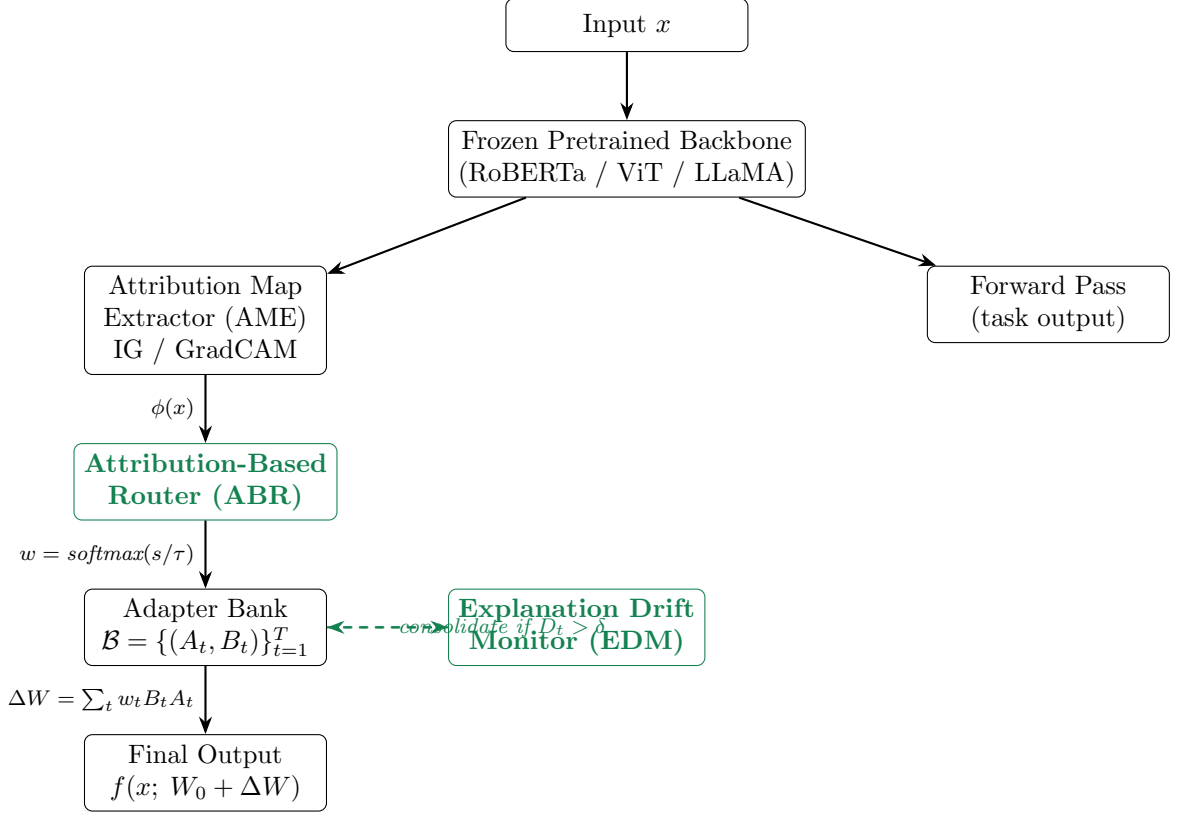


Figure 1: ESMCA architecture. Green modules are the novel contributions. Dashed arrows indicate training-time signals.

3.2 Component Descriptions

3.2.1 A. Task-Specific LoRA Adapter Bank

For each incoming task t , a low-rank adapter is trained:

$$\Delta W_t = B_t A_t, \quad B_t \in \mathbb{R}^{d \times r}, \quad A_t \in \mathbb{R}^{r \times k}, \quad r \in \{4, 8, 16\}$$

All previous adapters $\{(A_1, B_1), \dots, (A_T, B_T)\}$ are **frozen** when task $T + 1$ arrives. Only W_Q and W_V projections are adapted, following empirical best practice. Storage overhead is $< 2\%$ extra parameters per task.

3.2.2 B. Attribution Map Extractor (AME)

We use **Integrated Gradients (IG)** [3], which is differentiable, exact, and model-agnostic:

$$\phi_i(x) = (x_i - x'_i) \cdot \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha$$

where x' is a zero/padding baseline. An **attribution prototype** per task is computed over 256 validation samples:

$$\Phi_t = \frac{1}{|V_t|} \sum_{x \in V_t} \phi(x)$$

Only these compact prototype vectors are stored — **no raw training data is retained** (privacy-preserving).

3.2.3 C. Attribution-Based Router (ABR)

At inference, for a new input x , the attribution $\phi(x)$ is compared to all stored prototypes:

$$s_t = \cos(\phi(x), \Phi_t), \quad t = 1, \dots, T \quad (1)$$

$$w_t = \text{softmax}\left(\frac{s}{\tau}\right), \quad \tau = 0.1 \quad (2)$$

$$\Delta W_{\text{comp}} = \sum_{t=1}^T w_t \cdot B_t A_t \quad (3)$$

No task labels are required at inference. Routing is fully **explainability-driven**.

3.2.4 D. Explanation Drift Monitor (EDM)

After training on task $T + 1$, we measure attribution drift on held-out data from past tasks:

$$D_t = \text{KL}\left(\phi'(x_{\text{old}}) \parallel \Phi_t\right), \quad t < T + 1$$

If $D_t > \delta$ (a tunable threshold), forgetting is detected *before accuracy degrades*, triggering lightweight consolidation:

$$L_{\text{attrib}} = \sum_{t < T+1} \lambda \cdot \|\phi_{T+1}(x_t) - \Phi_t\|^2 \quad (\text{only when } D_t > \delta) \quad (4)$$

This produces the **Attribution Drift Score (ADS)** — a novel proactive forgetting metric.

3.3 Training Protocol

For each new task $T + 1$:

1. Initialize fresh LoRA adapter (A_{T+1}, B_{T+1})
2. Freeze all previous adapters $A_1, \dots, A_T$
3. Train on $\mathcal{D}_{T+1}$ with total loss:
$$L_{\text{total}} = L_{\text{task}} + \alpha \cdot L_{\text{attrib}}$$
4. Compute prototype Φ_{T+1} on validation set
5. Run EDM: compute D_t for all past tasks
6. If $D_t > \delta$: apply attribution-anchored regularization
7. Store $(A_{T+1}, B_{T+1}, \Phi_{T+1})$ in bank $\mathcal{B}$

4 Datasets

5 Baselines

6 Evaluation Metrics

6.1 Standard Continual Learning Metrics

- **ACC** — Average accuracy across all tasks after full sequential training

Table 2: Datasets used across domains for comprehensive evaluation.

Domain	Dataset	Tasks	Purpose
NLP	SuperGLUE	8 (BoolQ, RTE, WiC, CB...)	Standard diverse NLP CL benchmark
NLP	CLINC150	150 intents $\rightarrow$ 10 domains	Domain-shift routing stress test
NLP	MMLU	57 subjects (sequential)	Large-scale task-incremental stress test
Vision	Split CIFAR-100	10 tasks $\times$ 10 classes	Standard CL vision benchmark
Vision	DomainNet	6 domains (real, sketch...)	Visual domain-shift diversity
Vision	CORe50	11 sessions, 50 classes	Realistic lifelong visual recognition
Multimodal	VQA-v2 (splits)	Question-type boundaries	Cross-modal attribution routing

Table 3: Comparison baselines.

Method	Category
Sequential Fine-tuning	Lower bound (full catastrophic forgetting)
EWC	Regularization-based CL
O-LoRA	Orthogonal subspace LoRA
AM-LoRA	Attention-weighted adapter routing
L2R	Learned output-space routing
CABLE	Gradient-similarity routing
SEMA	Self-expanding mixture of adapters
Multi-task LoRA	Upper bound (joint training)

- **BWT** — Backward Transfer: $\text{BWT} = \frac{1}{T-1} \sum_{t=1}^{T-1} (R_{T,t} - R_{t,t})$, where $R_{i,j}$ is accuracy on task j after training on task i
- **FWT** — Forward Transfer: measures zero-shot reuse of past knowledge

6.2 Novel XAI-Specific Metrics (Contributions)

- **Attribution Drift Score (ADS)** — $\bar{D} = \frac{1}{T-1} \sum_t D_t$; lower values indicate less forgetting in attribution space
- **Routing Interpretability Score (RIS)** — Spearman correlation between routing weights w_t and ground-truth task labels; measures semantic meaningfulness without label supervision
- **Attribution Fidelity** — t-SNE cluster separability of task prototypes $\{\Phi_t\}$ in attribution space (qualitative + silhouette score)

7 Contributions

1. **First framework** to use attribution maps as the primary routing mechanism in continual learning — no task labels, no black-box router.
2. **Novel ADS metric** — proactive, interpretable forgetting detection before accuracy degrades.
3. **Privacy-preserving CL** — only compact attribution prototype vectors stored, never raw training data.

4. **Dual-use XAI** — the same attribution mechanism serves inference routing *and* training-time forgetting monitoring.
5. Systematic empirical study of explanation drift across tasks as a standalone finding.

8 Target Publication Venues

Venue	Track
NeurIPS / ICML / ICLR	Core method paper
EMNLP / ACL	NLP-focused evaluation
TMLR	Comprehensive multi-domain evaluation

References

- [1] McCloskey, M., & Cohen, N. J. (1989). Catastrophic interference in connectionist networks: The sequential learning problem. *Psychology of Learning and Motivation*, 24, 109–165.
- [2] Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., & Chen, W. (2022). LoRA: Low-rank adaptation of large language models. *ICLR 2022*.
- [3] Sundararajan, M., Taly, A., & Yan, Q. (2017). Axiomatic attribution for deep networks. *ICML 2017*.